One Card, One City: How a Piece of Plastic Connects Everything

- FORADAŞ: a single smart card for transit, restaurants and culture
- From raw material to a ledger entry, in the reference city of Marisa
- A technical walk-through for an English Translation and Interpreting class

Spoken script

Good morning, everyone. Today I will show you what really happens when someone taps a card on a bus. We will follow one person, Derya, through one day in the fictional city of Marisa. On the way we will open the card, the radio, the software and the money behind it.

The Problem: Many Cards, Many Wallets

- Each city or operator issues its own card, app or ticket
- Tourists and commuters carry several instruments
- A shared card needs shared rules, not just a shared plastic

Spoken script

Think about a visit to a new city. You buy a paper ticket for the bus, another for the metro, and you pay cash in the restaurant. A single card can remove this friction. But a single card is not magic; it needs agreements, standards and software behind it.

What Is Real, What Is Designed, What Is Simulated

- Standards (ISO/IEC 14443, EMV…) are real and cited
- The architecture is our project design
- Payments, approvals and integrations are simulated

Spoken script

Academic honesty matters. The standards I name are real documents. The architecture is our own design decision. Nothing here moves real money or claims a real permit.

Marisa: A Coastal City with a Real Network Logic

- Five districts, three lines, six main stops (fictional)
- Switch between 2D, 3D, transit network, district and street views
- The real FORADAŞ prototype also uses real İETT and OpenStreetMap stop data

Spoken script

This is Marisa, a fictional coastal city of about one million people. It has five districts: Old Town, Harbour, Uptown, Campus and Business Bay. Three lines connect them: bus twelve, tram T2 and metro M1. Choose a view, then tap a district to explore it.

Design Your Own Network for Marisa

- Place stops, draw lines, set the headway
- See coverage, line load and connectivity at once
- Six map views and a real-data check

Spoken script

Now you are the planner. Place stops where people live, draw lines between them and choose how often vehicles come. The result panel shows who is served and which lines are overloaded. Switch to 3D and street view to see what the numbers mean for people.

The Cast: People, Organisations and Places

- Derya, a graphic designer from Uptown (our traveller)
- Marisa Transit: the bus, tram and metro operator
- Kemer Sofrası: a restaurant in Old Town
- FORADAŞ Programme: the card programme operator

Spoken script

Every story needs characters. Derya is a designer who lives in Uptown. Marisa Transit runs the vehicles, and the FORADAŞ programme runs the card. Kemer Sofrası is the restaurant where she will have lunch.

Derya's Morning: From Home to the Validator

- Phone app: balance, nearby stops, suggested route
- Walk to stop 0417, live departure board
- Bus 12 arrives; she reaches the validator



Spoken script

It is 7:48 in the morning. Derya checks the app and sees her balance and a suggested route. She walks six minutes to the stop and watches the departure board. The bus arrives. Click once more to see the tap.

Inside 300 Milliseconds: Thirteen Steps

- Click through each step: where is the data, who decides, what comes back?
- Steps 1–8 happen on the bus; 9–12 need the backend
- Local versus backend is shown on every step



Spoken script

Now the exciting part. I will click one step at a time. Watch the data move from the card to the reader, from the vehicle to the cloud, and back. Notice the tags: some steps are local and some need the backend.

The Card as an Engineered Object

- ISO/IEC 7810 ID-1 format: 85.60 × 53.98 × 0.76 mm
- About 5 g, PVC or PET-G laminated, matt finish
- Tactile edge mark and high-contrast print for accessibility

Spoken script

The card is a small engineered product. It has the same size as a bank card, so it fits every wallet and card holder. Every choice has a reason: the matt finish resists scratches, and the notch helps blind users.

Card Anatomy: The Exploded View

- Five layers, one antenna, one chip module
- Click to separate and name each layer
- No battery: all power arrives through the antenna



Spoken script

Here the layers separate. On top is a clear overlay that protects the print. In the middle is the inlay: a copper antenna and the chip module. There is no battery; the antenna harvests the energy.

The Antenna: Energy and Data from a Magnetic Field

- Reader coil → alternating magnetic field at 13.56 MHz
- Card coil → induced voltage (Faraday induction) → chip power
- Data: reader modulates the field; card answers by load modulation



Spoken script

Here is the physics. The reader sends alternating current into its coil, which creates a changing magnetic field. The card coil sits in that field, so a small voltage appears in it. That voltage powers the chip and lets the card answer.

The Chip: More Than a Memory Tag

- Integrated circuit: processor, memory, crypto engine
- Secure element: tamper-resistant storage and operations
- A simple tag only answers with a fixed number; a smart card computes

Spoken script

The chip is a very small computer. It has a processor, memory and a cryptographic engine. A simple memory tag only sends back an identifier. A smart card can prove who it is without revealing its secret.

RFID, NFC and Contactless: Related but Not Identical

- RFID is the family; NFC is a standardised 13.56 MHz subset
- ISO/IEC 14443: proximity cards (about 10 cm)
- ISO/IEC 18092 and NFC Forum: peer-to-peer, tag types, phone modes

Spoken script

People often use these words as if they were the same. RFID means radio-frequency identification and covers many frequencies. NFC is a short-range 13.56 megahertz technology built on RFID standards. Our transit card uses the same radio layer as contactless bank cards.

Six Different "OK"s in One Tap

- NFC communication, card authentication, user authentication
- Payment authorisation, fare validation, settlement
- They are different processes with different owners

Spoken script

A green light looks like one thing. But there are six different questions behind it. Radio contact, card identity, user identity, money approval, fare rule and settlement are all different. Mixing them up is the most common mistake in this field.

From Raw Material to Finished Card: 14 Stages

- Fourteen manufacturing and issuing stages
- Each stage has a quality gate
- A card that fails is scrapped and traced by batch

Spoken script

A card starts as plastic sheets. The antenna and chip are added, then the layers are fused. Then comes printing, testing and secure personalisation. At the end, every card is traceable to a batch.

Four Identifiers, Four Different Jobs

- Card identifier (UID / printed number)
- Account identifier (in the backend)
- Credential (what the card presents)
- Cryptographic keys (what proves it)

Spoken script

A card carries several different identities. The printed number helps customer support. The account is a backend record. The keys live inside the chip and are never shown.

Card Lifecycle: New to Revoked

- New → Active → Suspended or Blocked → Replaced
- Lost card: blocked in seconds, balance kept in the account
- Revoked is permanent

Spoken script

A card can be new, active, suspended or blocked. If Derya loses it, she blocks it in the app. The balance stays with her account, so a new card receives it. A revoked card can never be used again.

Transport Hardware: Who Does What

- Validator: NFC reader, QR scanner, display, buzzer, controller
- Vehicle computer: GNSS, modem, router, local time
- Driver console, gates, vending machines, top-up kiosks, inspection devices

Spoken script

A bus carries a small network. The validator reads the card. The vehicle computer knows where the bus is and talks to the cloud. Gates, vending machines and kiosks do the same job in stations.

Inside a Validator

- NFC front end and antenna
- Microcontroller with secure storage and rules
- Modem path, time source and power supply

Spoken script

This is the heart of the bus. The NFC antenna and controller talk to the card. The microcontroller runs the rules and keeps a local log. The power supply and clock keep everything alive and in time.

Behind the Tap: Microservices and Events

- Small services, each with one job and its own data
- Synchronous for decisions, asynchronous for everything else
- One failure must not stop boarding

Spoken script

After the validator, the request meets a set of small services. The gateway guards the door. The fare engine and account service decide. Events carry the news to everyone else, so a failure of one helper does not stop the bus.

Never Charge Twice: Idempotency

- Each tap has a unique key
- The ledger refuses a second use of the same key
- A retry returns the first result

Spoken script

Networks fail in the middle of a request. The validator cannot know if the first request arrived. So it sends it again with the same key. The backend recognises the key and returns the first answer.

No Signal: The Bus Keeps Working

- Authenticate locally, apply cached rules, store securely
- Upload and reconcile when the signal returns
- Offline acceptance is a priced and capped risk

Spoken script

Tunnels and dead zones exist. The validator can still authenticate the card and apply local rules. It stores a signed record and shows a green tick. Later the backend deduplicates and reconciles.

Kemer Sofrası: A Place Worth Visiting

- Restored stone arcade in Old Town
- Open hearth, vaulted ceiling, seasonal menu
- Accepts FORADAŞ at the table

Spoken script

Derya steps off the bus and walks a hundred metres to Kemer Sofrası. It is a restored stone arcade with a blue-tiled sign. Inside there is a vaulted ceiling and an open wood-fired hearth. Choose each room to look around.

The Restaurant Payment, Fourteen Steps

- Authorisation happens in about a second
- Clearing and settlement come later
- Different application, keys and parties than the bus tap

Spoken script

Derya asks to pay with her card. The POS creates a payment request. She taps, the card proves itself, and the platform approves. The money follows later through clearing and settlement.

Ready-Made Tours and Guide Options

- Four ready tours built from the tour data
- Audio guide, local guide or self-paced
- Places, food prices and times from the data, rail fares marked as estimates

Spoken script

Not everyone wants to plan. These four tours are cut from the same data as the tour page. Pick a tour, pick how you want to be guided, and step through the stops. Everything is simulated: nothing is booked and no money moves.

Money Words: Who Does What

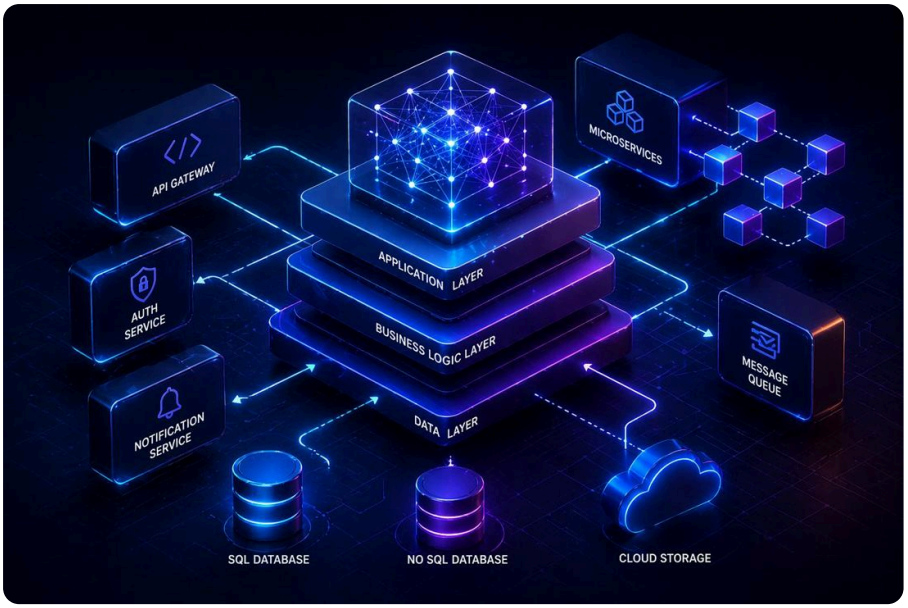
- Authorisation, capture, clearing, settlement, reconciliation
- Refund, reversal, chargeback
- Roles: issuer, acquirer, processor, network, merchant

Spoken script

Money words are easy to mix up. Authorisation is the permission. Clearing is the exchange of details. Settlement is the actual transfer of funds.

The Whole Platform in Five Layers

- Frontends → API → services → data → operations
- Click to reveal each layer
- Each layer has a single reason to exist



Spoken script

Let me zoom out to the whole system. There are five layers, from the apps at the top to monitoring at the bottom. I will reveal them one by one.

Network: From the Bus to the Data Centre

- VLANs, VPN, TLS, DNS, DHCP, firewalls
- Segmented networks: edge, application, data, management
- Redundant paths and monitored certificates

Spoken script

The bus is a mobile network node. Its traffic crosses the air, a VPN and a firewall. Inside the data centre the networks are separated. This limits the damage if something goes wrong.

Who Must Say Yes: The Stakeholder Map

- Permission is many agreements, not one stamp
- Not every organisation needs every approval
- Roles decide obligations: operator, issuer, processor, merchant

Spoken script

People ask: who gives permission for this? The honest answer is: many parties, in different ways. The municipality, the transport operator, a payment institution, suppliers and data protection authorities each have their role. Which approvals you need depends on what role your organisation plays.

A Compliance Roadmap, from Plan to Launch

- Eight stages from owner to launch
- Compliance is continuous, not a single approval
- Certification is evidence, not a substitute for compliance

Spoken script

Here is a possible roadmap. It starts with an owner and ends with launch and service levels. In the middle sit the payment decision, licences, security and a pilot.

Standards: What They Are and Whether They Are Mandatory

- Technical standards, certification frameworks, laws
- Mandatory in a scenario or an engineering recommendation
- Check the current edition before citing

Spoken script

Standards have different statuses. ISO 14443 is a technical standard for the radio. PCI DSS is an industry framework you must follow if you handle card data. ISO 27001 is a management system you can be certified against.

Privacy by Design: Collect Less, Protect More

- Collect only what the service needs (data minimisation)
- Token instead of name on devices and in analytics
- Short retention, strict access, rights respected

Spoken script

Travel history reveals a lot about a person. So we collect only what we need. The validator never sees Derya's name; it sees a token. Analytics use aggregated counts only.

Security Scenarios: Attack, Detect, Contain, Recover

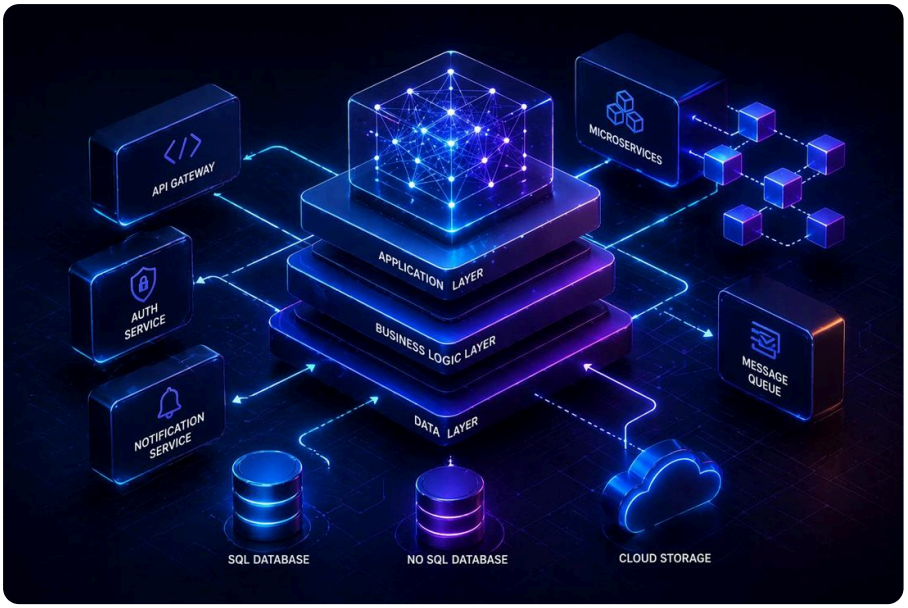
- Defensive scenarios only
- Prevention plus detection plus response
- Authentication, authorisation, accounting are three different things

Spoken script

Let us look at security as a story. Each scenario has an attacker goal, a detection method, a limit on the damage and a recovery. Notice that prevention alone is never enough.

Troubleshooting Lab: How a Technician Finds the Fault

- Choose a fault: card, reader, network, clock
- Symptoms → checks → finding → fix
- Every step has a reason



Spoken script

Now a technician arrives. First we see what the passenger sees and what the logs say. Then step by step the technician tests one component at a time. Finally we fix it and confirm.

The Failure Catalogue

- Six families, each with symptoms, logs, causes and fixes
- Detailed diagnostic flow exists for four faults in the lab
- Prevention matters as much as repair

Spoken script

Faults come in families. Cards, readers, networks, back-end services, payments and people. For each family we plan detection, repair, escalation and prevention.

Support Workflow: From Complaint to Closure

- Eight stages from report to closure
- Verification without exposing personal data
- Incidents feed prevention

Spoken script

After launch, operations matter every day. Derya reports a problem, a ticket opens, support verifies, diagnoses, escalates and resolves. She is told the result, and the lesson goes into prevention.

Dashboards for Four Audiences (Simulated Data)

- Operator, merchant, municipality, system administrator
- All numbers are simulated
- Analytics are aggregated; individuals are protected

Spoken script

Different people need different views. The operator watches validators and failures. The restaurant watches payments. The municipality sees aggregated trends, never individuals.

Who Pays Whom, and Why

- Costs: cards, hardware, software, cloud, support, compliance
- Revenue: card sales, merchant fees, transport service fees, data-free insights
- All figures in this project are illustrative

Spoken script

A system needs money to run. Cards, readers, servers, support and audits all cost money. The revenue may come from small merchant fees, a service fee from the operator and card sales. We use illustrative numbers only.

Same Card in Another City: What Must Be Agreed

- Standardise the radio and message formats
- Keep tariffs, currencies and privacy rules local
- Interoperability is an agreement, not a feature

Spoken script

Can Derya use the card abroad? Only if the two programmes agreed. The standards help, but contracts, currency and law matter as much.

More Uses, Each with Its Own Agreement

- Metro, tram, ferry, bike share, parking
- Museums, universities, libraries, events
- Each integration needs hardware, software and legal arrangements

Spoken script

The card can grow into many services. But each service needs its own readers, rules, agreements and security review. One card does not automatically open every door.

Conclusion: One Card, Many Systems, One Careful Design

- A tap is a radio event, a software decision and a financial promise
- Reliability comes from design, operations and honest limits
- Terminology precision is part of technical accuracy

Spoken script

Let us go back to Derya's tap. It was a magnetic field, a cryptographic proof, a fare decision and a ledger entry. Behind it are standards, contracts, people and careful engineering. And in translation, choosing the right word is part of that precision. Thank you.

Sources and Verification Status

- Standards cited by series; check the current edition
- Legal references need verification of the current text
- No source here is invented

Spoken script

These are the primary sources for this presentation. I cite standards by series, since editions change. Every legal point must be checked against the current text.